Session 19->24: Final Project
UART IP verification by using UVM

Final Project



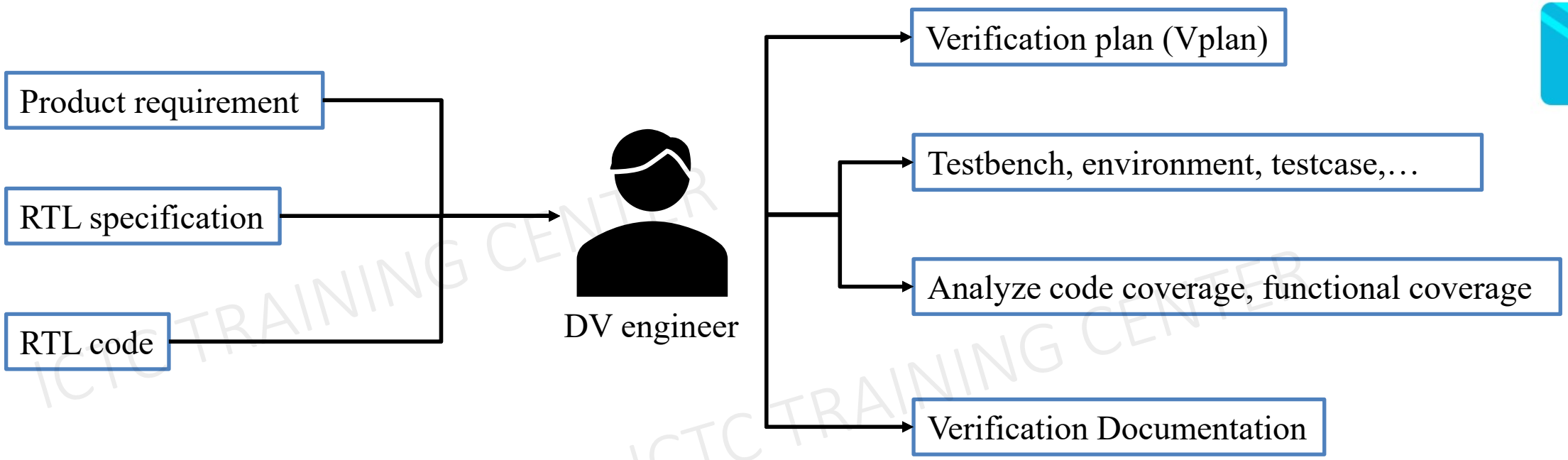
We finally reach here!!! Awesome work !!!
Let’s do the last practice - the most challenge and interesting thing in this course. This project base on the real project!!!

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THE FINAL PROJECT – UART IP Verification

RECAP: DV Engineer

What is DV engineer responsibility?



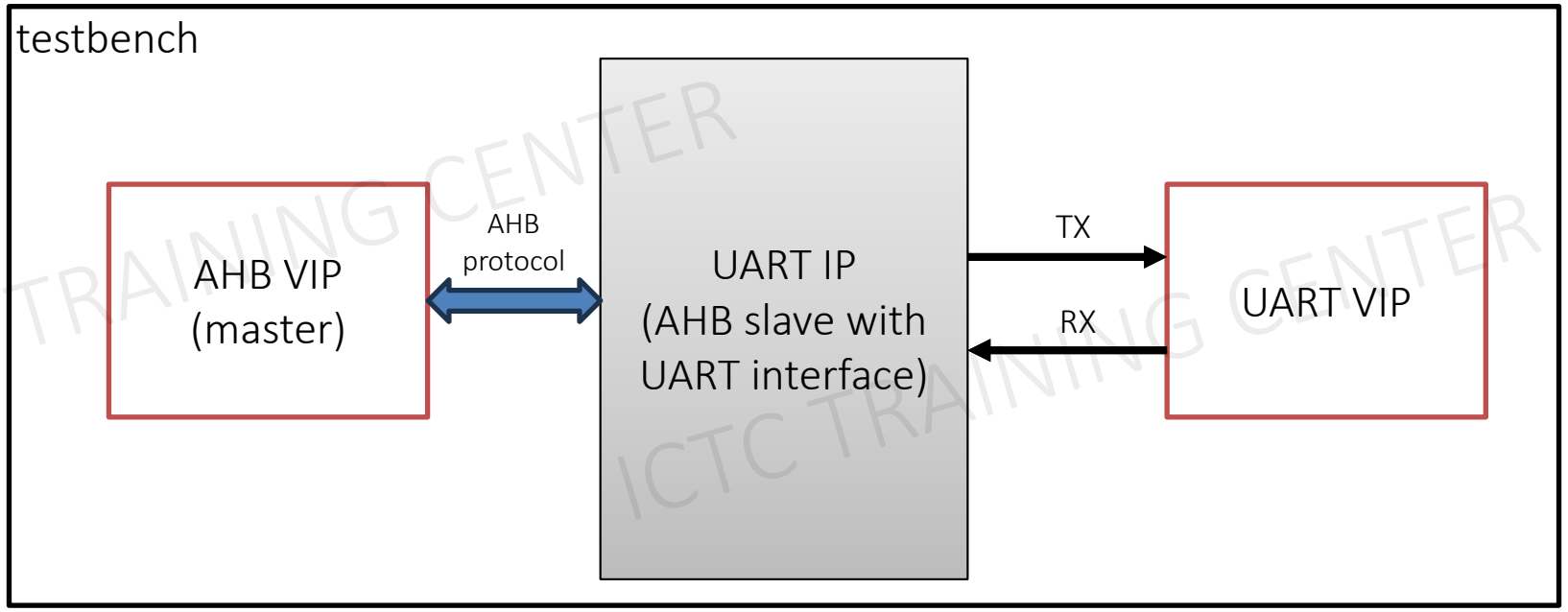
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In fact, DV engineer will be join in vary stage of chip production

- Part from RTL simulation (SoC & IP level). In SoC level, DV engineer also do:
  - UPF simulation
  - AMS simulation
  - GLS simulation: Pre-layout, Post-layout (Worst case, Best case,…) and PG netlist
- Support FPGA team to debug C code.
- Support CHIP testing.
- Develop new DV methodology (script, optimize working, structural environment and improve database)

Final project: UART IP verification

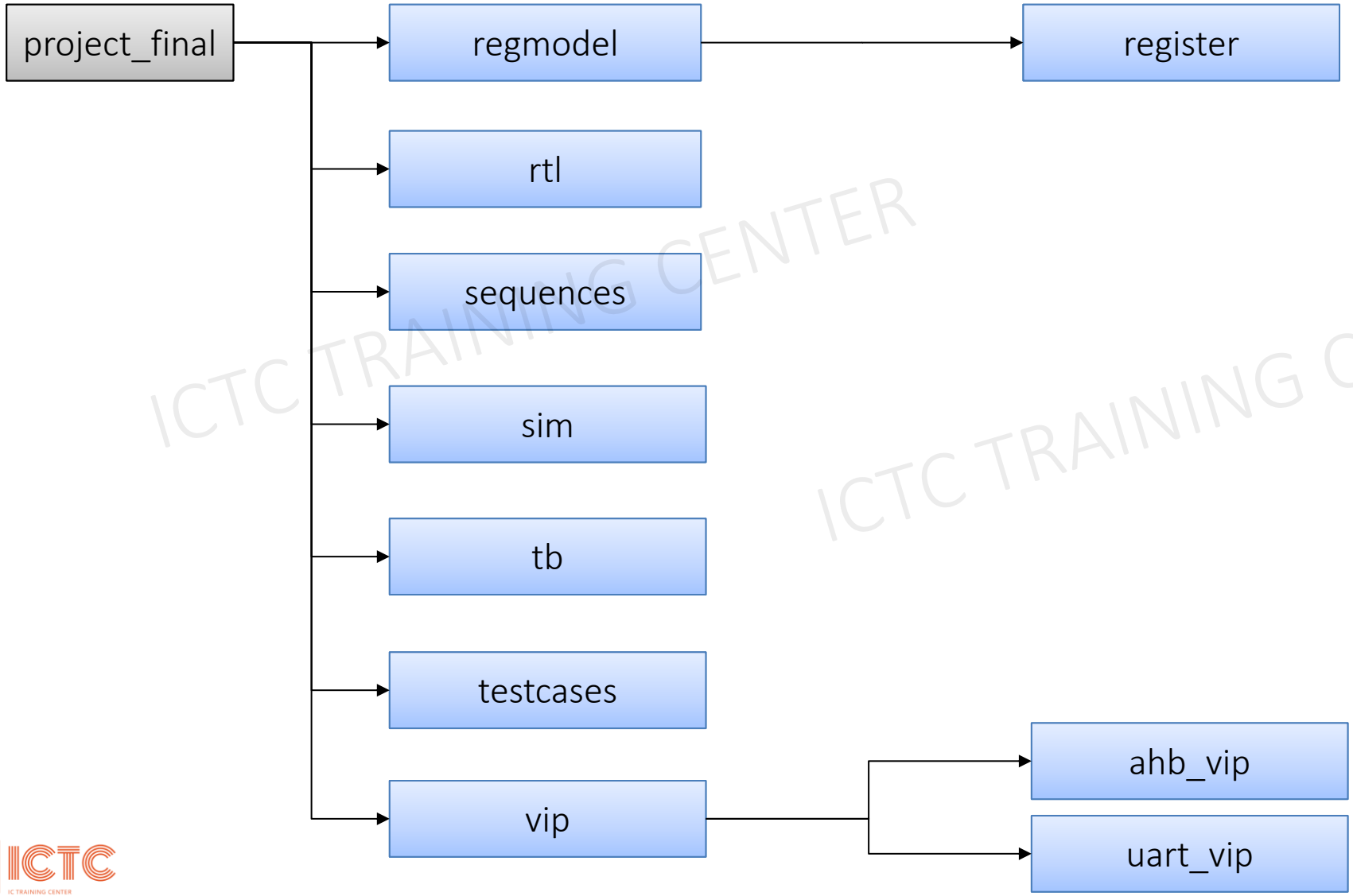
Testbench Overview



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Final project: UART IP verification

Directory structure



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Final project: UART IP verification

Requirement

Instructions

- **Preparation:**
  1. Reference UART IP RTL specification: IP specification
  2. Copy /ictc/student_data/share/dv_advanced/project_final to your working directory
- **Your task:**
  1. Investigate UART IP specification.
  2. Estimate your planning to complete this project.
  3. Create your verification plan.
  4. Draw testbench structure in detail.
  5. Build testbench, environment, scoreboard,..
  6. Create test scenario and checker according to Vplan.
  7. Implement functional coverage

- **Project requirement completion (Max grade is 6):**
  - Complete step 1 -> step 6
  - 50% testcase created with checker and data comparision in scoreboard
  - Passing rate is 100%

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THANK YOU